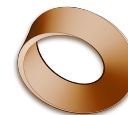


L4 – Building Series and Parallel Circuits

P2 Electricity • AQA Combined Science: Trilogy 8464



Physics 11 R • Mr. Rawson • 10 students

11 R Spec	Mon 31 Aug, P6, 15:05–15:55 P2.2 — aP2-18 to aP2-22, aP2-24	Next Lesson	Tue 8 Sep, P4 — 8 days 4 of P2 (sequence provisional)
Equations	$V = IR$ — RECALL , already secure from the 28 Aug review; and $R_{total} = R_1 + R_2$ (series) — RECALL . Parallel total resistance is never calculated : AQA Combined Science excludes it explicitly (aP2-21). Qualitative only.		
Kit	Per pair (5 pairs): power supply or cell pack, leads, two lamps, ammeter, voltmeter. [[ALEX: decide — what is actually in the PH1 cupboard: power packs or cell holders? separate ammeters/voltmeters or multimeters? enough for 5 sets? Ask Joel. The handout assumes cell + 2 lamps + 1 ammeter + 1 voltmeter per pair, moved between positions.]]		
To print	Per student (10): the L4 handout and the homework sheet.		
Homework	Six questions, 21 marks, graduated: $V = IR$ applied to a series circuit, then per-branch in parallel, then two qualitative explanations. Due Tue 8 Sep.		

By the end of the lesson students can:

- build a series and a parallel circuit from a circuit diagram, with the ammeter in the loop and the voltmeter across the component;
- state the current and potential difference rules for each arrangement, *from their own measurements*;
- apply $V = IR$ to a real component using measured values, and $R_{total} = R_1 + R_2$ to a series pair.

Timing

Time	Phase	What happens
0–5	Do Now	Three $V = IR$ one-liners on the board — find I , find V , find R . This is the 28 Aug review cashing in; it should be quick. [[ALEX: decide — pitch depends on how far the PhET task actually got on 28 Aug. If they built parallel circuits in the sim, open by naming that; if they only got as far as series, the parallel half below is genuinely new.]]
5–12	The two arrangements	Series and parallel on the board, standard symbols. Name the distinction: one loop vs branches. Then the hook — <i>predict</i> : unscrew one lamp in each circuit; what happens to the other? Collect predictions, do not resolve. The circuits will.
12–30	Circuit 1 — series	Pairs build from the handout diagram. Current at position 1, then move the ammeter to position 2; pd across each lamp and the supply. Circulate hunting for the two wiring errors: voltmeter in the loop (kills the circuit), ammeter across a component (shorts it). Then they unscrew a lamp and test their prediction.
30–42	Circuit 2 — parallel	Rebuild to the second diagram. Current at positions 1, 2, 3; pd across each branch. The two findings that matter: branch currents <i>add</i> to the supply current, and every branch sees the <i>full</i> supply pd. Unscrew a lamp again — the other stays lit.
42–48	The rules	Consolidation table on the handout, filled in from their own numbers, not dictated. Resolve the predictions from minute 5 out loud. State the resistance rules: series adds; parallel — qualitative only, more paths means less opposition.
48–50	Wrap	Set homework (due Tue 8 Sep). Pack-down discipline: this is P6, last period — leads coiled, lamps back in trays, or it costs you the morning.

Questions to ask

1. The ammeter reads the same either side of the lamp. Where did the current “go” inside the lamp — and what does that tell you current actually is?
2. Your two lamp pds add up to the supply pd. Will that still be true if the lamps are different? Why?
3. In the parallel circuit, which lamp is brighter — and what does the voltmeter say about why?
4. I add a third lamp in series. What happens to the current, and why? Now in parallel instead — what happens to the current *from the supply*?
5. Why are the lights in this room wired in parallel and not in series?

Common misconceptions

- **“Current gets used up.”** The single biggest P2 misconception. Position 1 vs position 2 in the series circuit exists to kill it with their own readings. If the two readings differ, it is a loose connection — say so before someone concludes the misconception is data.
- **Voltmeter wired into the loop.** The circuit dies (voltmeter resistance is huge). Catch it while circulating; it is the most common build error.
- **“Voltage flows.”** Pd is *across*, current is *through*. Police the language all lesson; AQA credits either “potential difference” or “voltage” but never “voltage through”.
- **“More resistors always means more resistance.”** True in series only. The parallel circuit’s rising supply current is the counter-evidence — draw the extra loop and count paths.
- **Trying to calculate parallel R_{total} .** A keen student may know the reciprocal formula from elsewhere. Praise it, then say clearly that Combined Science never asks it (aP2-21) — qualitative comparison only.

Differentiation

Support: pair the shakier builders with confident ones (10 students, 5 pairs — choose the pairs, do not let them self-select). A photographed working series circuit on the board as a wiring crib once the first pair succeeds.

Extend: a two-lamps-in-series branch inside the parallel circuit (a combined circuit, aP2-18); predict all three ammeter readings with $V = IR$ *before* measuring, then check.

After the lesson

- **11R’s prior coverage is still genuinely unknown** — the inherited status sheet’s 11R column is blank, and the 28 Aug board is the only confirmed ground. If today exposes gaps below P2.2 (symbols, $Q = It$), log them; do not assume the 11Q hypothesis held.
- **Next is Tue 8 Sep (P4), then the Fri 11 Sep P3+P4 double** — the natural slot for RP15 (wire length + series/parallel combinations) if today’s build skills are sound. Decide after marking the homework.
- If the “current gets used up” idea survived contact with the ammeter, open Tue 8 Sep with it rather than moving on.
- Note which kit actually worked — dead lamps, dodgy leads — for Joel, before the Fri double depends on it.
- Mark the Class Log entry **Taught**; homework due Tue 8 Sep.